19. For the system of equations
$x+2y+3z=1,\quad 2x+y+3z=2$
$5x+5y+9z=4$
(a) there is only one solution
(b) there exists infinitely many solutions
(c) there is no solution
(d) None of these

20. If $|A| \neq 0$, then A is
(a) non- singular matrix
(b) singular matrix
(c) zero matrix
(d) none of these

21. The system of linear equations $x+y+z=2,\ 2x+y-z=3, 3x+2y+k=4$ has unique solution, if k is not equal to
(a) 4 (b) 0
(c) –4 (d) 1

22. Cramer's rule is applicable only to get an unique solution when
(a) $\Delta_z \neq 0$ (b) $\Delta_x \neq 0$
(c) $\Delta \neq 0$ (d) $\Delta_y \neq 0$

23. If $\frac{a_1}{x}+\frac{b_1}{y}=c_1, \frac{a_2}{x}+\frac{b_2}{y}=c_2,$

$\Delta_1=\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}, \Delta_2=\begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix} \Delta_3=\begin{vmatrix} c_1 & a_1 \\ c_2 & a_2 \end{vmatrix}$

then (x, y) is
(a) $\left(\frac{\Delta_2}{\Delta_1}, \frac{\Delta_3}{\Delta_1}\right)$ (b) $\left(\frac{\Delta_3}{\Delta_1}, \frac{\Delta_2}{\Delta_1}\right)$
(c) $\left(\frac{\Delta_1}{\Delta_2}, \frac{\Delta_1}{\Delta_3}\right)$ (d) $\left(\frac{-\Delta_1}{\Delta_2}, \frac{-\Delta_1}{\Delta_3}\right)$

24. If $|A_{n\times n}| = 3$ and $|adjA| = 243$ then the value n is
(a) 4 (b) 5
(c) 6 (d) 7

25. Rank of a null matrix is
(a) 0 (b) –1
(c) ∞ (d) 1

Miscellaneous problems

1. Find the rank of the matrix
$A=\begin{pmatrix} 1 & -3 & 4 & 7 \\ 9 & 1 & 2 & 0 \end{pmatrix}$.

2. Find the rank of the matrix
$A=\begin{pmatrix} -2 & 1 & 3 & 4 \\ 0 & 1 & 1 & 2 \\ 1 & 3 & 4 & 7 \end{pmatrix}$.

3. Find the rank of the matrix
$A=\begin{pmatrix} 4 & 5 & 2 & 2 \\ 3 & 2 & 1 & 6 \\ 4 & 4 & 8 & 0 \end{pmatrix}$.

4. Examine the consistency of the system of equations:
$x+y+z=7$, $x+2y+3z=18$, $y+2z=6$

5. Find k if the equations $2x+3y-z=5$, $3x-y+4z=2$, $x+7y-6z=k$ are consistent.

6. Find k if the equations $x+y+z=1$, $3x-y-z=4$, $x+5y+5z=k$ are inconsistent.

7. Solve the equations $x+2y+z=7$, $2x-y+2z=4$, $x+y-2z=-1$ by using Cramer's rule.

8. The cost of 2kg. of wheat and 1kg. of sugar is ₹100. The cost of 1kg. of wheat and 1kg. of rice is ₹80. The cost of 3kg. of wheat, 2kg. of sugar and 1kg of rice is ₹220. Find the cost of each per kg., using Cramer's rule.

9. A salesman has the following record of sales during three months for three items A, B and C, which have different rates of commission.

Months	Sales of units			Total commission drawn (in ₹)
	A	B	C	
January	90	100	20	800
February	130	50	40	900
March	60	100	30	850

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